KUET · ESE 3207 · Solar Photovoltaic Systems

ESE 3207 Question Map

Every question from the 2019, 2020, 2021, 2022, 2023 term papers, grouped by topic and ranked by how often it comes back and how many marks it carries. Answers are in the ESE 3207 Study Pack (HTML and PDF) next to this file.

24-hour list | Pass plan | Frequency grid | Section A | Section B

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 4 h for Part A and 2 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×5, 2022 ×4, 2021 ×3, 2020 ×2, 2019 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Retake tip: read the Concept Primer first (30 minutes). Part B is the easier half: on-grid/off-grid and net metering (B5), temperature and irradiance effects (B1), bypass diodes and shading (B2) and the solar home system sizing numerical (B6) come every year. In Part A, learn the I-V curve, FF and efficiency (A4, with a calculator), losses and the S-Q limit (A5) and quartzite-to-wafer (A6). Junction physics numericals (A3) are the hardest: use the calculator to practise them last.

Section A: Solar cell physics and technology

Block 1: very likely and easy · 1.2 h

Topic	Chance	Level	Covered
A6 Silicon wafer and c-Si cell production: quartzite to wafer, cell fabrication, lithography, TCO, supporting layers	Very likely 100%	Easy	18%

A5 Losses and efficiency limits: optical and electrical losses, design rules, back surface field, thermodynamic and Shockley-Queisser limits	Very likely 100%	Easy	31%
A4 Solar cell parameters and equivalent circuit: I_{SC}, V_{OC}, FF, efficiency, R_s and R_{sh}, quantum efficiency	Very likely 100%	Easy	42%

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
A8 Multi-junction, perovskite and third-generation cells; generations of solar cells	Very likely 100%	Medium	58%
A1 Semiconductor basics: bands, doping, Fermi level, direct/indirect gap, drift and diffusion, Poisson and continuity equations	Very likely 100%	Medium	70%
A7 Thin-film cells: III-V, CdTe, CIGS, thin-film silicon (a-Si:H), materials criteria	Very likely 87%	Medium	82%
✔ Pass line: 82% of Section A marks covered, enough to choose 3 of the 4 questions (about 3.5 h)			
A2 Generation and recombination: radiative, SRH, Auger, surface; diffusion length; photocurrent numerical	Very likely 87%	Medium	90%

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
A3 Junctions: p-n junction solar cell, p-i-n, metal-semiconductor, diode equation; junction numericals	Very likely 100%	Hard	100%

Section B: PV systems

Block 1: very likely and easy · 1.7 h

Topic	Chance	Level	Covered
B5 PV system types: standalone (SHS), grid-connected, with storage, hybrid; net metering; grid interfacing; standalone applications; water pumping; SHS installation and testing	Very likely 100%	Easy	25%
B1 PV cell, module, panel and array; series and parallel; effects of irradiance, temperature, area, orientation and tilt	Very likely 100%	Easy	45%
B7 Economics and ecology: payback time, energy payback time, LCOE, life-cycle cost, types of cost, carbon footprint, recycling	Very likely 100%	Easy	58%
B2 Shading and mismatch; hot spots; bypass and blocking diodes	Very likely 93%	Easy	67%

Block 2: very likely, medium effort · 1.5 h

Topic	Chance	Level	Covered
B6 Designing and sizing a PV system (solar home system numericals)	Very likely 100%	Medium	78%
✔ Pass line: 78% of Section B marks covered, enough to choose 3 of the 4 questions (about 2.4 h)			
B3 Battery, charge controller (set points, PWM, shunt and series regulators), MPPT (P&O, incremental conductance), DC-DC converters	Very likely 87%	Medium	96%

Only if time is left · 0.4 h

Topic	Chance	Level	Covered
B4 Inverters: types by waveform and by mode, THD, converter efficiency and inverter selection numericals	Possible 47%	Easy	100%

Pass plan

Five papers: 2019, 2020, 2021, 2022 and 2023. Normal papers: 3 hours, 210 marks, answer any THREE of four questions in each section (35 marks each). 2020 was a short exam (1.5 h, 120 marks, any TWO of three questions per section; its Section B also held the wafer and lithography questions). Section A is solar cell physics and technology (Rahaman sir, MR: Parts I-III, following Smets et al. *Solar Energy*) and Section B is PV systems (Muhit sir, MMM: lectures L1-L11, following Smets and Messenger *Photovoltaic Systems Engineering*). This pack sorts by **topic**, not question number. **Every topic is in the pass set** except thin-film cells (A7) and multi-junction/perovskite/third generation (A8), which are the next to add; they still come up every year, so learn at least the multi-junction and CdTe answers. In the paper: Section A answer Q1 (semiconductors, recombination), Q2 (junction, parameters, R_s/R_{sh}) and Q3 (losses, SQ limit, wafer); Section B answer Q5 (module, temperature, shading), Q6 (charge controller and SHS design numerical, 15-20 marks) and Q7 (on-grid/off-grid, net metering). Answers follow the teachers' slides first and the two books they cite; each answer says where a book was used. Every numerical was re-checked in Python.

SECTION A: SOLAR CELL PHYSICS AND TECHNOLOGY

SECTION B: PV SYSTEMS

6 topics

cover 73% of all Section A marks asked

- Semiconductors, doping, Fermi level, transport (A1); generation, recombination and the absorption numerical (A2)
- p-n and p-i-n cells, Schottky contact, V_{bi}/W numericals (A3); parameters, FF numericals, equivalent circuit, R_s/R_{sh} (A4)
- Losses, design rules, BSF, SQ limit (A5); quartzite to wafer, cell fabrication, lithography, TCO (A6)
- Next to add: thin film (A7); multi-junction, perovskite and third generation (A8)

7 topics

cover 100% of all Section B marks asked

- Cell, module, array; series and parallel; irradiance, temperature, tilt (B1)
- Shading, hot spots, bypass and blocking diodes, mismatch (B2)
- Charge controller, set points, PWM, MPPT, DC-DC (B3); inverters, THD and numericals (B4)
- System types, grid interfacing, net metering, pumping (B5); SHS sizing numericals (B6); economics and ecology (B7)

- **Section A Q1 every year (semiconductor physics):** doping with band diagrams and Fermi level; direct vs indirect gap; SRH, Auger, radiative and surface recombination; Fermi level, Poisson, continuity, diffusion length (A1, A2).
- **Section A Q2 every year (the cell as a device):** p-n and p-i-n working principle; external parameters I_{sc}, V_{oc}, FF, η; equivalent circuit; effect of R_s and R_{sh}; V_{bi}/W numerical (A3, A4).
- **Section A Q3 (losses and manufacturing):** types of loss, design rules, BSF, thermodynamic and Shockley-Queisser limits; quartzite → Si wafer, cell fabrication, lithography, TCO (A5, A6).
- **Section A Q4 (advanced cells):** multi-junction working and challenges, lattice matching of GaInP/GaAs/Ge, CdTe layer structure, III-V and thin-film Si, perovskite, generations (A7, A8).
- **Section B Q5-Q6 (module to controller):** cell/module/array, series-parallel, irradiance and temperature effects with a temperature numerical; shading, hot spots, bypass and blocking diodes, mismatch; charge controller and set points, PWM, MPPT (P&O, IncCond), DC-DC converters (B1-B3).
- **Section B Q6-Q8 (systems):** SHS sizing numerical (2019 = 2023, 15-20 marks: always attempt), inverter types and THD, on-grid/off-grid/with storage, grid interfacing

conditions, net metering and Bangladesh eligibility, EPBT vs payback, LCOE, LCC, carbon footprint and recycling (B4-B7).

- **Always draw:** band diagrams (doped, junction, Schottky), recombination paths, p-n and p-i-n cell, I-V curve with MPP, equivalent circuit, R_s/R_{sh} curves, SQ curve, wafer flow chart, CdTe and triple-junction stacks, series-parallel I-V, shaded string with bypass diode, set-point plot, MPPT P-V curve, buck/boost circuit, inverter waveforms, on-grid/off-grid block diagrams, net meter.

Frequency grid

Marks asked from each topic in each paper. Darker means more marks. Click a topic to jump to its questions.

Marks per paper: 1–8 9–16 17–24 25+

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION A: SOLAR CELL PHYSICS AND TECHNOLOGY: MARKS ASKED PER PAPER

Topic	2019	2020	2021	2022	2023	Years	Marks	
A6 Silicon wafer and c-Si cell production: quartzite to wafer, cell fabrication, lithography, TCO, supporting layers	35	36	23	30	12	5/5	136	Pass
A5 Losses and efficiency limits: optical and electrical losses, design rules, back surface field, thermodynamic and Shockley-Queisser limits	25	15	18	22	23	5/5	103	Pass
A4 Solar cell parameters and equivalent circuit: I_{sc}, V_{oc}, FF, efficiency, R_s and R_{sh}, quantum efficiency	10	18	30	8	24	5/5	90	Pass
A8 Multi-junction, perovskite and third-generation cells; generations of solar cells	7	29	40	23	20	5/5	119	Next to a
A1 Semiconductor basics: bands, doping, Fermi level, direct/indirect	15	23	25	11	25	5/5	99	Pass se

gap, drift and diffusion, Poisson and continuity equations								
A7 Thin-film cells: III-V, CdTe, CIGS, thin-film silicon (a-Si:H), materials criteria	28		17	27	15	4/5	87	Next to a
A2 Generation and recombination: radiative, SRH, Auger, surface; diffusion length; photocurrent numerical	20		25	11	10	4/5	66	Pass s
A3 Junctions: p-n junction solar cell, p-i-n, metal-semiconductor, diode equation; junction numericals	10	25	12	18	11	5/5	76	Pass

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION B: PV SYSTEMS: MARKS ASKED PER PAPER

Topic	2019	2020	2021	2022	2023	Years	Marks	
B5 PV system types: standalone (SHS), grid-connected, with storage, hybrid; net metering; grid interfacing; standalone applications; water pumping; SHS installation and testing	60	6	16	32	36	5/5	150	Pass se
B1 PV cell, module, panel and array; series and parallel; effects of irradiance, temperature, area, orientation and tilt	26	8	29	40	18	5/5	121	Pass se
B7 Economics and ecology: payback time, energy payback time, LCOE, life-cycle cost, types of cost, carbon footprint, recycling	6	18	18	23	14	5/5	79	Pass se
B2 Shading and mismatch; hot spots; bypass and blocking diodes		16	10	10	17	4/5	53	Pass s
B6 Designing and sizing a PV system (solar home system numericals)	15	12	13	6	20	5/5	66	Pass set

B3 Battery, charge controller (set points, PWM, shunt and series regulators), MPPT (P&O, incremental conductance), DC-DC converters	33		20	22	35	4/5	110	Pass set
B4 Inverters: types by waveform and by mode, THD, converter efficiency and inverter selection numericals			17	7		2/5	24	Pass

Section A: Solar cell physics and technology

topics A1-A8 (questions 1-4; 2020: questions 1-3 plus 4(a)-(b))

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

A6 Silicon wafer and c-Si cell production: quartzite to wafer, cell fabrication, lithography, TCO, supporting layers

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 136 marks in total

STUDY THESE

- Quartzite (SiO_2) + carbon in an arc furnace → metallurgical Si (98-99%) → HCl → trichlorosilane → distillation → Siemens CVD → polysilicon (9N+) → Czochralski / float-zone ingot → wire saw → etch, polish
- Cell process: damage etch, texturing, $POCl_3$ diffusion (n^+ emitter), PSG removal, edge isolation, SiN_x ARC (PECVD), screen-print Ag/Al, co-firing (Al-BSF), testing
- Lithography: oxide, photoresist, mask + UV exposure, develop, etch, strip; used for fine contacts and PERL cells
- TCO: transparent ($E_g > 3$ eV) and conductive oxide (ITO, SnO_2:F, ZnO:Al) used as front electrode on thin-film cells; supporting layers (substrate/superstrate, buffer, window, ETL/HTL, back reflector)

> 2023 Q3(d) 12 marks
>
> Discuss the production of silicon wafer from quartzite.

> 2022 Q2(a) 13 marks
>
> What are the key characteristics of quartzite that make it a suitable source material for silicon wafer production? Discuss the production of silicon wafer from quartzite.

> 2022 Q4(a) 10 marks
>
> Why is TCO used in a solar cell? Also explain why III-V material is considered for thin-film solar technology. (III-V part: A7)

> 2022 Q2(d) 7 marks
>
> Why are supporting layers used in solar cells and what considerations are taken into account when selecting this crucial component?

> 2021 Q2(c) 12 marks
>
> Briefly explain the production of c-Si wafer from quartzite.

> 2021 Q2(b) 11 marks
>
> What is meant by lithography? Write down the steps of lithography with a proper diagram.

> 2020 Q4(b) 18 marks
>
> Describe in detail how to make a polished wafer of Si from the starting materials.

> 2020 Q4(a) 12 marks
>
> What is meant by lithography? Write down the steps of lithography with a proper diagram.

> 2020 Q3(c) 6 marks
>
> Why are supporting layers used in solar cells? Discuss.

> 2019 Q3(a) 14 marks
>
> Discuss the production of silicon wafer from quartzite.

> 2019 Q3(b) 14 marks
>
> Discuss the fabrication of a c-Si solar cell from a silicon wafer.

> 2019 Q3(c) 7 marks
>
> What is TCO? Explain.

Full answer: ESE 3207 Study Pack, topic A6.

A5 Losses and efficiency limits: optical and electrical losses, design rules,

Pass set · Very likely · 100% · Easy

back surface field, thermodynamic and Shockley-Queisser limits

Asked in 5 of 5 papers · 103 marks in total

STUDY THESE

- Optical losses: shading, reflection, parasitic absorption, transmission; spectral mismatch: non-absorption ($E_{ph} < E_g$) and thermalisation ($E_{ph} > E_g$)
- Electrical losses: bulk, surface and contact recombination; ohmic (series) and shunt losses
- Design rules: texture + ARC, thin fingers, back reflector (light management); passivation, BSF, heavy doping under contacts, L > base thickness (recombination management); low R_s
- Limits: thermodynamic (Carnot-type, ~86% for full concentration with infinite junctions; ~68% one sun); SQ detailed-balance limit for one junction ~33.7% at $E_g \approx 1.34$ eV; Si ~29.4% with Auger

> 2023 Q3(a) 10 marks
> Discuss the different types of losses in a solar cell.

> 2023 Q3(b) 6 marks
> Why is the back surface field important in a high-efficiency solar cell? Give a proper diagram.

> 2023 Q3(c) 7 marks
> Briefly explain the thermodynamic limit and the Shockley-Queisser limit.

> 2022 Q3(c) 9 marks
> Briefly explain the design rules for a solar cell with respect to loss.

> 2022 Q1(c) 13 marks
> Discuss in detail the balance limit of solar cell efficiency according to the Shockley-Queisser limit.

> 2021 Q4(b) 10 marks
> Briefly explain the efficiency losses of a solar cell; define IQE and EQE.

> 2021 Q3(a) 8 marks
> “Performance of a solar cell depends on irradiance.” Justify the statement. (answered in B1)

> 2020 Q2(b) 15 marks

> How can transparency loss and excess excitation loss be minimized in a solar cell? Explain in detail. What are the main challenges in multi-junction solar cells? (MJ part: A8)

> `2019` `Q1(c)` `10 marks`
>
> Discuss the different types of losses in a solar cell.

> `2019` `Q2(c)` `15 marks`
>
> Discuss in detail the balance limit of solar cell efficiency according to the Shockley-Queisser limit.

Full answer: ESE 3207 Study Pack, topic A5.

A4 Solar cell parameters and equivalent circuit: I_{sc}, V_{oc}, FF, efficiency, R_s and R_{sh}, quantum efficiency

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 90 marks in total

STUDY THESE

- STC: 1000 W/m^2, AM1.5 spectrum, cell temperature 25 °C
- I_{sc} (current at V = 0), $V_{oc} = (kT/q)\ln(J_{ph}/J_0 + 1)$, $FF = V_m I_m/(V_{oc} I_{sc})$, $\eta = V_{oc} I_{sc} FF/(G \cdot A)$
- Real cell: $I = I_L - I_0[e^{q(V+IR_s)/nkT} - 1] - (V + IR_s)/R_{sh}$; R_s should be small, R_{sh} large; both lower FF
- EQE = carriers collected / incident photons; IQE = carriers collected / absorbed photons; $EQE = (1 - R)(1 - T) \cdot IQE$

> `2023` `Q2(b)` `12 marks`
>
> Briefly explain external solar cell performance parameters with proper equations.

> `2023` `Q2(c)` `12 marks`
>
> Briefly explain the effect of series and parallel resistance on the I-V characteristics of a solar cell.

> `2022` `Q3(b)` `8 marks`
>
> Draw and briefly explain the equivalent circuit of a realistic solar cell which considers all the losses, and also write the modelled equation.

> `2021` `Q5(d)` `11 marks`

> A solar panel 61 cm × 46 cm at STC: V_{oc} = 21.5 V, I_{sc} = 3.21 A, V_{MPP} = 20 V, I_{MPP} = 2.5 A. Calculate the fill factor and efficiency of this panel. (asked in Section B)

> `2021` `Q4(a)` `9 marks`
> Draw an equivalent circuit of an ideal solar cell and a solar cell with series resistance (R_s) and shunt resistance (R_p), mentioning their modelled equations.

> `2021` `Q4(b)` `10 marks`
> Briefly explain the efficiency losses of a solar cell. Also define internal quantum efficiency and external quantum efficiency. (losses part: A5)

> `2020` `Q1(a)` `6 marks`
> A solar cell 12 cm × 12 cm at STC: V_{oc} = 0.8 V, I_{sc} = 3 A, V_{MPP} = 0.75 V, I_{MPP} = 2.5 A. Calculate the fill factor and efficiency.

> `2020` `Q1(c)` `12 marks`
> With the help of mathematical equations, explain the influence of series and shunt resistance on the performance of a solar cell.

> `2019` `Q2(b)` `10 marks`
> Draw and discuss the equivalent circuit of an ideal solar cell and an actual solar cell with R_s and R_p.

Full answer: ESE 3207 Study Pack, topic A4.

A8 Multi-junction, perovskite and third-generation cells; generations of solar cells

Next to add · Very likely · 100% · Medium

Asked in 5 of 5 papers · 119 marks in total

STUDY THESE

- Multi-junction: stacked cells with falling E_g (GaInP 1.86 / GaAs 1.42 / Ge 0.67 eV), each absorbs its part of the spectrum; series-connected through tunnel junctions; record > 47%
- Challenges: bandgap matching, current matching (series current = lowest cell), lattice matching (mismatch → dislocations), tunnel junctions, cost
- Lattice mismatch $f = (a_{layer} - a_{sub})/a_{sub}$; Ge, GaAs, GaInP all have a ≈ 5.65 Å
- Perovskite ABX_3 ($MAPbI_3$): tunable gap 1.2-2.3 eV, strong absorption, long diffusion length, solution-processed, > 26% single, > 33% on Si tandem; problems: stability, lead

- Generations: I c-Si wafers; II thin films; III concepts beyond SQ (MJ, hot carrier, IB, MEG, spectral conversion, concentrators) and new materials

> `2023` `Q4(a)` `10 marks`
>
> What is a multijunction solar cell? Explain the working principle of a multijunction solar cell with a proper diagram.

> `2023` `Q4(d)` `10 marks`
>
> Briefly explain third-generation solar cell technology with its significance.

> `2022` `Q4(c)` `7 marks`
>
> Explain the challenges for multijunction solar cells.

> `2022` `Q2(c)` `8 marks`
>
> What is lattice mismatch? Justify the use of Ge, GaAs and GaInP to form triple-junction cells.

> `2022` `Q4(d)` `8 marks`
>
> What distinguishes perovskite solar cells from traditional silicon-based cells in terms of their construction and materials?

> `2021` `Q4(c)` `10 marks`
>
> What are the approaches to increase the efficiency of a solar cell? Explain how a multi-junction solar cell increases efficiency compared to a single-junction solar cell.

> `2021` `Q3(c)` `10 marks`
>
> Same question as 2022 2(c).

> `2021` `Q7(a)` `14 marks`
>
> What are the properties and characteristics of perovskite solar cells? Write down the classifications of perovskite solar cells. Explain the basic structure of perovskite solar cells. (asked in Section B)

> `2021` `Q4(d)` `6 marks`
>
> Write a short note on generations of solar cells.

> `2020` `Q2(b)` `15 marks`
>
> Second part: main challenges in multi-junction solar cells.

> `2020` `Q3(a)` `14 marks`
>
> Explain the working principle of high-efficiency perovskite solar cells.

> `2019` `Q4(b)` `7 marks`
>
> Same question as 2022 2(c).

Full answer: ESE 3207 Study Pack, topic A8.

A1 Semiconductor basics: bands, doping, Fermi level, direct/indirect gap, drift and diffusion, Poisson and continuity equations

Pass set · Very likely · 100% · Medium

Asked in 5 of 5 papers · 99 marks in total

STUDY THESE

- Band model: valence band (bound electrons), conduction band (free electrons), bandgap E_g (Si 1.12 eV); a broken bond = free electron + hole
- Doping: group V donor (P) → n-type, E_F near E_C; group III acceptor (B) → p-type, E_F near E_V; $n = N_D$, $p = n_i^2/N_D$
- Fermi level = energy with 50% occupation probability; $f(E) = 1/(1 + e^{(E-E_F)/kT})$; $n = N_C e^{-(E_C-E_F)/kT}$
- Transport: drift $J = q(n\mu_n + p\mu_p)E$ and diffusion $J = qD\,dn/dx$; Einstein $D/\mu = kT/q$; Poisson $d^2\psi/dx^2 = -\rho/\varepsilon$; continuity $\partial n/\partial t = (1/q)\partial J_n/\partial x + G - R$

> 2023 Q1(a) 10 marks
>
> What is the purpose of doping in a semiconductor? Explain different types of doping with band diagram levelling the Fermi level energy.

> 2023 Q1(c) 15 marks
>
> Write the physical significance of the Poisson equation, continuity equation, minority carrier diffusion length, density of states and Fermi level energy, including their equations.

> 2022 Q1(a) 11 marks
>
> Define Fermi level in a semiconductor. Briefly explain the physical significance of the Poisson and continuity equations together for semiconductor material.

> 2021 Q1(a) 10 marks
>
> Same question as 2023 1(a).

> 2021 Q1(b) 15 marks
>
> Briefly explain direct and indirect band gap semiconductors. Describe the SRH and Auger recombination process mechanisms with their band diagrams. (recombination part: A2)

> 2020 Q2(a) 11 marks
>
> Define doping. Discuss the p-n junction's working procedure of a solar cell. (p-n part: topic A3)

> 2020 Q1(b) 12 marks
>
> Explain the basic two transport mechanisms in a semiconductor device.

> 2019 Q1(a) 5 marks
>
> Why is the Fermi level of an n-type semiconductor near the lowest conduction band energy, and for a p-type near the highest valence band energy? What is a direct and indirect bandgap semiconductor?

> 2019 Q1(b) 10 marks
>
> What are the four main ideas represented in quantum theory? Explain surface recombination, and mention its significance in a solar cell.

Full answer: ESE 3207 Study Pack, topic A1.

A7 Thin-film cells: III-V, CdTe, CIGS, thin-film silicon (a-Si:H), materials criteria

Next to add · Very likely · 87% · Medium

Asked in 4 of 5 papers · 87 marks in total

STUDY THESE

- Thin film = absorber a few µm thick on glass/foil, needs a direct or strong absorber; low material and energy use
- III-V (GaAs, InP, GaInP, AlGaAs): direct gap, very high absorption and efficiency, radiation hard, tunable gap and lattice; expensive (space, concentrators)
- CdTe: superstrate glass/TCO/n-CdS/p-CdTe/back contact; E_g 1.45 eV direct; cheapest thin film (First Solar)
- a-Si:H: hydrogen passivates dangling bonds; low diffusion length so p-i-n drift device; Staebler-Wronski degradation; tandem with nc-Si:H

> 2023 Q4(b) 10 marks
>
> Explain the operation of a CdTe solar cell using its layer structure.

> 2023 Q4(c) 5 marks
>
> Why is III-V technology for solar cells called thin-film solar cell?

> 2022 Q4(b) 10 marks
>
> Explain the operation of a CdTe solar cell using its layer structure.

> 2022 Q4(a) 10 marks
>
> Why TCO is used; why III-V material is considered for thin-film technology.

> 2022 Q2(b) 7 marks
>
> What are the various types of materials used in the construction of thin-film solar cells and how do they contribute to the overall efficiency?

> 2021 Q2(a) 12 marks
>
> Mention the names of some III-V materials used in thin-film solar cell production. Also discuss the design of thin-film silicon solar cells.

> 2021 Q3(d) 5 marks
>
> What are the criteria for materials to be used in solar cells?

> 2019 Q4(a) 3 marks
>
> Mention the names of some III-V materials used in thin-film solar cell production.

> 2019 Q4(d) 15 marks
>
> Discuss the design of thin-film silicon solar cells.

> 2019 Q4(c) 10 marks
>
> Why is hydrogen used in thin-film silicon alloys? Why is a thin-film silicon solar cell called a drift device?

Full answer: ESE 3207 Study Pack, topic A7.

A2 Generation and recombination: radiative, SRH, Auger, surface; diffusion length; photocurrent numerical

Pass set · Very likely · 87% · Medium

Asked in 4 of 5 papers · 66 marks in total

STUDY THESE

- Generation: a photon with $E \geq E_g$ lifts an electron to the CB (electron-hole pair); recombination is the reverse; in equilibrium G = R and $np = n_i^2$
- Radiative (band-to-band, gives a photon; direct-gap materials), SRH (via a trap level from defects/impurities; dominant in Si), Auger (three carriers, energy to a third carrier; high doping), surface (dangling bonds)
- Minority-carrier diffusion length $L = \sqrt{(D\tau)}$: average distance a minority carrier travels before recombining; must exceed the absorber thickness

- Lambert-Beer: $\Phi(x) = (1-R)\Phi_0 e^{-\alpha x}$; photon flux $\Phi_0 = P\lambda/(hc)$; J_{ph} = q × absorbed flux

> `2023` `Q1(b)` `10 marks`
>
> Explain the recombination process through SRH and Auger recombination with proper diagrams.

> `2022` `Q1(b)` `11 marks`
>
> Briefly explain the role of carrier generation and recombination in maintaining dynamic equilibrium. Define minority carrier diffusion length with physical significance.

> `2021` `Q1(b)` `15 marks`
>
> Direct and indirect band gap; SRH and Auger recombination with band diagrams.

> `2021` `Q1(c)` `10 marks`
>
> The total absorption in a d = 300 µm thick c-Si wafer for light of wavelength 500 nm. Incident photons reflected from the surface are 0.38 and absorption coefficient $\alpha = 1.11 \times 10^4$ cm^{-1}. The incident irradiance is 1000 W/m^2. Calculate the photocurrent density.

> `2019` `Q2(a)` `10 marks`
>
> Explain Auger recombination in detail.

> `2019` `Q1(b)` `10 marks`
>
> Surface recombination and its significance in a solar cell (second part).

Full answer: ESE 3207 Study Pack, topic A2.

A3 Junctions: p-n junction solar cell, p-i-n, metal-semiconductor, diode equation; junction numericals

Pass set · Very likely · 100% · Hard

Asked in 5 of 5 papers · 76 marks in total

STUDY THESE

- p-n junction: electrons and holes diffuse across, leaving fixed ions (depletion region) and a built-in field; light-made pairs reaching the field are separated: e^- to n, h^+ to p
- $V_{bi} = (kT/q)\ln(N_A N_D/n_i^2)$; $W = \sqrt{[(2\varepsilon/q)(V_{bi} \pm V)(1/N_A + 1/N_D)]}$; $E_{max} = \sqrt{[(2q/\varepsilon)V_{bi} N_A N_D/(N_A+N_D)]}$

- p-i-n: thick intrinsic layer between thin p and n; field across the whole i-layer: drift device (a-Si:H, perovskite)
- Metal-semiconductor: Schottky (rectifying) if $\phi_m > \phi_s$ for n-type; ohmic if $\phi_m < \phi_s$ (reversed for p-type); diode $I = I_0(e^{qV/kT} - 1)$

> 2023 Q2(a) 11 marks
>
> Explain the working principle of a p-i-n junction solar cell using a schematic diagram.

> 2022 Q3(a) 10 marks
>
> Explain the working principle of a p-n junction solar cell using a schematic diagram.

> 2022 Q3(d) 8 marks
>
> A c-Si wafer is doped with 10^{16} acceptor atoms/cm^3. A 1 µm thick emitter layer is formed at the surface with a uniform 10^{18} donors/cm^3. Assume a step p-n junction, all dopants ionized, $n_i = 1.5 \times 10^{10}$ cm^{-3} at 300 K. Calculate the built-in voltage, width of the depletion region and maximum electric field at the metallurgical junction.

> 2021 Q3(b) 12 marks
>
> Same wafer as 2022 3(d). Calculate the electron and hole concentrations in the p- and n-type quasi-neutral regions at thermal equilibrium.

> 2020 Q2(a) 11 marks
>
> Define doping. Discuss the p-n junction's working procedure of a solar cell.

> 2020 Q2(c) 4 marks
>
> Write the diode equation and draw the I-V curve of an ideal diode.

> 2020 Q3(b) 10 marks
>
> Si cell at 300 K: A = 1.5 cm^2, $N_A = 5 \times 10^{17}$, $N_D = 8 \times 10^{17}$ cm^{-3}, $D_n = 22$, $D_p = 10$ cm^2/s, $\tau_n = 3 \times 10^{-7}$ s, $\tau_p = 5 \times 10^{-7}$ s, generation rate $G_L = 10^{22}$ cm^{-3}s^{-1}. Calculate the photocurrent. The cell is reverse biased with 2 V.

> 2019 Q1(d) 10 marks
>
> Explain the metal-semiconductor junction with energy band diagrams under different bias conditions.

Full answer: ESE 3207 Study Pack, topic A3.

Section B: PV systems

topics B1-B7 (questions 5-8; 2020: questions 5-6)

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

B5 PV system types: standalone (SHS), grid-connected, with storage, hybrid; net metering; grid interfacing; standalone applications; water pumping; SHS installation and testing

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 150 marks in total

STUDY THESE

- Standalone: PV + charge controller + battery + (inverter) + loads; types: DC load only, DC + control, + battery, AC/DC with inverter
- Grid-connected: PV + grid-tied inverter + meter; no battery (grid is the storage); with battery (bimodal inverter, backup); hybrid: PV + diesel/wind with electronic control
- Net metering: bi-directional meter, bill on (import − export); Bangladesh Net Metering Guidelines 2018
- Grid interfacing: voltage and frequency within limits, phase/frequency synchronisation, anti-islanding, power quality (THD, DC injection), protection, power factor

> 2023 Q7(a) 14 marks
>
> Discuss the working principle of on-grid and off-grid solar PV systems with proper diagrams.

> 2023 Q7(b) 14 marks
>
> What is net metering? How does it work? Explain with proper illustration.

> 2023 Q8(a) 8 marks
>
> Provide examples of real-life applications of standalone PV systems and explain how they meet specific energy needs.

> 2022 Q7(a) 12 marks
>
> Discuss the working principle of an on-grid solar PV system with storage.

> 2022 Q5(a) 10 marks
>
> What are the key components of a solar home system and explain the role of each component in converting sunlight into usable electric energy.

> 2022 Q7(b) 10 marks
>
> What is net metering? How does it work?

> 2021 Q8(a) 8 marks
>
> How does a grid-connected PV system work? Write the conditions for grid interfacing of an on-grid PV system.

> 2021 Q7(b) 8 marks
>
> What is net metering? Write down the eligibility conditions for a customer to be considered for a net metering connection in Bangladesh.

> 2020 Q5(c) 6 marks
>
> Write down the components of an on-grid PV system.

> 2019 Q7(a) 10 marks
>
> Discuss the working principle of on-grid and off-grid solar systems with necessary diagrams.

> 2019 Q7(c) 6 marks
>
> What are the conditions for grid interfacing of an on-grid PV system?

> 2019 Q7(b) 11 marks
>
> What is net metering? How does it work?

> 2019 Q7(d) 8 marks
>
> Write down the eligibility conditions for a customer to be considered for a net metering connection in Bangladesh.

> 2019 Q5(d) 9 marks
>
> Explain the working principle of a PV-powered water pumping system.

> 2019 Q6(b) 6 marks
>
> Write the merits and demerits of a PV pumping system when it is directly connected to the PV array or to a battery.

> 2019 Q8(d) 10 marks
>
> Discuss the testing, installation and replacing procedures of a solar home system with different components.

Full answer: ESE 3207 Study Pack, topic B5.

B1 PV cell, module, panel and array; series and parallel; effects of irradiance, temperature, area, orientation and tilt

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 121 marks in total

STUDY THESE

- Cell (~0.5-0.6 V) → module (36/60/72 cells in series) → string (modules in series, V adds) → array (strings in parallel, I adds)
- Irradiance: $I_{sc} \propto G$, V_{oc} rises only logarithmically; power roughly $\propto G$
- Temperature: V_{oc} falls ~2.3 mV/°C per Si cell, I_{sc} rises slightly, P_{max} falls 0.4-0.5 %/°C: $P(T) = P_{STC}[1 + \gamma(T - 25)]$
- Area: current $\propto$ area, voltage independent of area; orientation: face south (Bangladesh), tilt ≈ latitude, output $\propto \cos$(incidence angle)

> 2023 Q5(a) 12 marks
>
> What are the effects of radiation and PV module temperature on PV cell performance? Explain with necessary figures.

> 2023 Q5(b) 6 marks
>
> Write short notes on (i) PV module, (ii) PV array.

> 2022 Q8(b) 10 marks
>
> In what ways does temperature affect solar cell efficiency? What strategies and technologies can be employed to reduce temperature-related challenges in PV systems?

> 2022 Q5(c) 7 marks
>
> The maximum rated power of a crystalline solar cell is 75 Wp. Calculate the maximum power output at 45, 55 and 65 °C cell temperature. Decrease in maximum power per °C rise (from STC 25 °C) is −0.45 %/°C.

> 2022 Q7(c) 13 marks
>
> Discuss the effect of PV module temperature, orientation and tilt angle on the performance of a solar PV system.

> 2022 Q5(b) 10 marks
>
> How does the surface area of a solar cell impact its output current, and what key factors should be considered in optimizing the design of a solar cell for efficient energy conversion?

> 2021 Q3(a) 8 marks
>
> “Performance of a solar cell depends on irradiance.” Justify the statement. (asked in Section A)

> 2021 Q6(d) 7 marks
>
> How does heat affect solar panel efficiency? Describe.

> 2021 Q5(a) 6 marks
>
> Write short notes on (i) PV cell, (ii) PV module, (iii) PV array.

> 2021 Q5(c) 8 marks
>
> Explain the working principle of photovoltaic panels in series and parallel connection. Why are series and parallel connections necessary in PV systems?

> 2020 Q5(a) 8 marks
>
> “Performance of a solar cell depends on irradiance.” Justify the statement.

> 2019 Q8(a) 10 marks
>
> Same question as 2022 7(c).

> 2019 Q5(a) 4 marks
>
> Write short notes on (i) PV module and (ii) PV array.

> 2019 Q5(b) 12 marks
>
> Same question as 2021 5(c).

Full answer: ESE 3207 Study Pack, topic B1.

B7 Economics and ecology: payback time, energy payback time, LCOE, life-cycle cost, types of cost, carbon footprint, recycling

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 79 marks in total

STUDY THESE

- Payback time = initial cost / annual saving (money); energy payback time EPBT = energy invested / annual energy yield (1-3 years for modern modules)
- LCOE = $\Sigma(I_t + M_t + F_t)/(1+r)^t \div \Sigma E_t/(1+r)^t$; grid parity and socket parity
- Life-cycle cost = present worth of all costs: $PW = C \cdot x^n$ with $x = (1+i)/(1+d)$; recurring: $Pa = (1-x^n)/(1-x)$

- Carbon footprint: CO_2 from making modules (mostly Si purification) vs CO_2 avoided: PV ~20-50 g CO_2/kWh vs coal ~800-1000; recycling glass, Al, Si, Ag; hazards: Pb, Cd, landfill

> 2023 Q7(c) 7 marks
>
> What are the differences between payback time and energy payback time?

> 2023 Q8(c) 7 marks
>
> Does a solar panel reduce carbon footprint and how does production of solar energy contribute to carbon footprint?

> 2022 Q5(d) 8 marks
>
> Why is energy payback time important in PV system design? What factors influence the energy payback time of a PV system?

> 2022 Q8(c) 7 marks
>
> Same question as 2023 8(c).

> 2022 Q8(d) 8 marks
>
> Can solar modules be reused after reaching the end of their life cycle, and if so, how? What environmental impact is associated with improper disposal of solar modules?

> 2021 Q8(d) 10 marks
>
> How does the concept of levelized cost of electricity work? Describe.

> 2021 Q6(a) 8 marks
>
> What is the carbon footprint of making solar panels? Explain the environmental impact of PV systems.

> 2020 Q6(b) 5 marks
>
> Why is energy payback time important in PV system design?

> 2020 Q6(c) 13 marks
>
> Compare the life-cycle cost of a highway construction warning sign that is PV powered vs a gasoline generator. 24 h/day operation, load 3 kWh/day, 20-year life. PV: 750 W array at \$4/W, \$900 of batteries replaced every 5 years, \$300 charge controller, maintenance \$100/year. Generator: 500 W for \$250, about 545 gallons of gasoline per year, maintenance about \$1500/year, replaced every 5 years. Inflation 3%, discount rate 10%.

> 2019 Q6(c) 6 marks
>
> Discuss the different types of cost of a PV power generation system.

Full answer: ESE 3207 Study Pack, topic B7.

B2 Shading and mismatch; hot spots; bypass and blocking diodes

Pass set · Very likely · 93% · Easy

Asked in 4 of 5 papers · 53 marks in total

STUDY THESE

- Shading: static (buildings, trees) and dynamic (clouds, dust, snow); in a series string the shaded cell limits the current of the whole string
- Hot spot: shaded cell is driven into reverse bias and dissipates the power of the good cells as heat
- Bypass diode: in parallel with a cell group, reverse polarity; conducts only when its group is shaded; I-V becomes a staircase with several power peaks
- Blocking diode: in series between array and battery; stops the battery discharging into the array at night (and protects parallel strings)
- Mismatch: cells/modules with different I_{sc}, V_{oc} (manufacturing tolerance, shading, soiling, temperature, ageing, orientation); fixes: sorting/binning, bypass diodes, string inverters, module-level MPPT (optimisers, micro-inverters)

> 2023 Q5(c) 6 marks
>
> Why are bypass and blocking diodes used in a solar panel?

> 2023 Q5(d) 11 marks
>
> What are the different types of mismatches in solar power systems?

> 2022 Q6(a) 10 marks
>
> How does mismatch in current affect the performance of series-connected PV modules, and what strategies or technologies are employed to mitigate current mismatch issues to optimize overall efficiency?

> 2021 Q5(b) 10 marks
>
> Describe the PV system's shading effect. Give a justifiable explanation of how the PV system performs when there is shading.

> 2020 Q5(b) 16 marks
>
> Explain shading of a PV system. With proper justification, explain the performance of a PV system during shading.

Full answer: ESE 3207 Study Pack, topic B2.

B6 Designing and sizing a PV system (solar home system numericals)

Pass set · Very likely · 100% · Medium

Asked in 5 of 5 papers · 66 marks in total

STUDY THESE

- 1 Load table: daily energy E = Σ (number × watt × hours) Wh/day; connected load = Σ (number × watt)
- 2 PV array: $W_p = E/(\eta_{bat} \times PSH)$ (optionally × 1.3 for dust, wiring, temperature); number of modules = W_p/module rating
- 3 Battery: Ah = E × days of autonomy/(V_{bat} × DOD)
- 4 Charge controller: current ≥ 1.25 × array current (I_{sc} or W_p/V); inverter VA ≥ 1.25 × connected load/pf
- 5 Switch rating ≥ 1.25 × circuit current; wire size A = 2L I ρ/ΔV (ΔV ≤ 3%)

> 2023 Q6(c) 20 marks
>
> SHS load profile: (i) laptop 2 × 20 W, 3 h/day; (ii) lamp 4 × 30 W, 8 h/day; (iii) colour TV 2 × 50 W, 6 h/day; (iv) 1 × 800 W, 2 h/day (geyser). Inverter pf unity, DOD 60%, autonomy 4 days, battery charging efficiency 75%, peak sun hours 5, battery 12 V. Design a PV module. Also calculate (i) battery capacity, (ii) charge controller, (iii) DC to AC inverter capacity, (iv) switch capacity, (v) wire size.

> 2022 Q6(d) 6 marks
>
> Design a solar PV system for a house with 3 fans of 70 W each running 4 h/day, 3 tube lights of 35 W each running 8 h/day, and a refrigerator of 250 W running 12 h/day (compressor runs 6 h on and 6 h off). Battery autonomy 2 days, DOD 50%, battery charging efficiency 80%, peak sun hours 6, battery voltage 12 V.

> 2021 Q7(c) 13 marks
>
> How to design and size solar photovoltaic systems, and installation of a solar PV system.

> 2020 Q6(a) 12 marks
>
> Write down the design guidelines of all components of a 3 kW off-grid PV system.

> 2019 Q6(d) 15 marks
>
> Same load profile and data as 2023 6(c).

Full answer: ESE 3207 Study Pack, topic B6.

B3 Battery, charge controller (set points, PWM, shunt and series regulators), MPPT (P&O, incremental conductance), DC-DC converters

Pass set · Very likely · 87% · Medium

Asked in 4 of 5 papers · 110 marks in total

STUDY THESE

- Charge controller: prevents overcharge and deep discharge, blocks reverse current, load control, low-voltage disconnect
- Set points: VR (regulation), VRH (regulation hysteresis), LVD (low-voltage disconnect), LVDH (LVD hysteresis); basic parameters: nominal system voltage, load current, PV array current
- PWM: full current until VR, then pulses whose duty cycle falls as SOC rises; shunt regulator shorts the array, series regulator opens it
- MPPT: DC-DC converter + algorithm that matches load to source impedance; P&O (perturb, observe dP) vs incremental conductance (dI/dV = −I/V at MPP)
- DC-DC: buck $V_o = DV_{in}$, boost $V_o = V_{in}/(1-D)$, buck-boost; parameters: input/output voltage range, current/power rating, efficiency, switching frequency, ripple

> 2023 Q6(a) 7 marks
>
> Why is a charge controller needed in a solar PV system?

> 2023 Q6(b) 8 marks
>
> Illustrate the key set points of a charge controller and explain their behaviour using a labelled diagram.

> 2023 Q8(b) 10 marks
>
> What is the function of an MPPT device and how does it ensure maximum power transfer from a PV module?

> 2023 Q8(d) 10 marks
>
> What are the basic parameters of a DC to DC converter? Write the basic working principles of DC-DC conversion of a solar PV system.

> 2022 Q6(b) 12 marks
>
> What role does a charge controller play in a solar system? How does the incremental conductance algorithm function in MPPT charge controllers and what advantages does it offer over simpler algorithms like perturb and observe?

> 2022 Q8(a) 10 marks
>
> Same question as 2023 8(d).

> 2021 Q6(b) 12 marks
>
> What is considered the most important feature in a charge controller? How do solar MPPT charge controllers work? Explain in detail.

> 2021 Q8(c) 8 marks
>
> Same question as 2023 8(d).

> 2019 Q8(b) 10 marks
>
> Why is a charge controller important in a solar PV system? Write the operation principle of the shunt regulator and series regulator.

> 2019 Q8(c) 5 marks
>
> What are the basic parameters of a charge controller?

> 2019 Q5(c) 10 marks
>
> Explain the basic working principle of a PV solar battery charging technique using the PWM method.

> 2019 Q6(a) 8 marks
>
> Write short notes on (i) specific gravity, (ii) autonomy days, (iii) luminous efficiency, and (iv) total harmonic distortion (THD).

Full answer: ESE 3207 Study Pack, topic B3.

B4 Inverters: types by waveform and by mode, THD, converter efficiency and inverter selection numericals

Pass set · Possible · 47% · Easy

Asked in 2 of 5 papers · 24 marks in total

STUDY THESE

- Inverter converts DC to AC; functions: power conversion, voltage regulation, frequency control (+ MPPT, synchronisation, anti-islanding for grid-tied)
- By waveform: square (cheap, THD ~48%), modified sine (stepped, ~25-30%), pure sine (THD < 3-5%); by mode: standalone, grid-tied, bimodal/hybrid
- THD = $\sqrt{(V_2^2 + V_3^2 + \ldots)}/V_1$; reduce with PWM/multilevel switching, LC filters, higher switching frequency, pure-sine design
- Efficiency $\eta = P_{out}/P_{in}$; loss = $P_{in}(1 - \eta)$; inverter size ≥ total simultaneous load / η (with ~25% margin)

> 2022 Q6(c) 7 marks
>
> A power converter has a specified input power of 300 W and a manufacturer-specified efficiency of 88%. How much is the power loss within the converter? What is the output power?

> 2021 Q6(c) 8 marks
>
> A house has 3 × 60 W lights, 2 × 90 W fans, 1 × 60 W radio, AC loads rated 230 V. All loads are powered simultaneously. A 12 V battery is available. Choose the appropriate inverter, assuming inverter efficiency 90%.

> 2021 Q8(b) 9 marks
>
> What is total harmonic distortion in solar inverters? How can total harmonic distortion in inverters be reduced? Discuss the different types of solar inverters based on output waveform.

Full answer: ESE 3207 Study Pack, topic B4.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
a-Si:H	hydrogenated amorphous silicon
Ah	ampere-hour
ALCC	annualised life-cycle cost
AM	air mass
ARC	anti-reflection coating
ARV	array reconnect voltage
AZO	aluminium-doped zinc oxide
BOS	balance of system
BSF	back surface field
CB	conduction band
CIGS	copper indium gallium diselenide
CZ	Czochralski
CZTS	copper zinc tin sulfide

DOD	depth of discharge
EPBT	energy payback time
EQE	external quantum efficiency
ETL	electron transport layer
EVA	ethylene vinyl acetate (encapsulant)
FA	formamidinium
FF	fill factor
FTO	fluorine-doped tin oxide
FZ	float zone
HIT	heterojunction with intrinsic thin layer
HTL	hole transport layer
HVD	high-voltage disconnect
IBC	interdigitated back contact
IEEE	Institute of Electrical and Electronics Engineers
IncCond	incremental conductance
IQE	internal quantum efficiency
ITO	indium tin oxide
LCA	life-cycle assessment
LCC	life-cycle cost
LCOE	levelized cost of electricity
LRV	load reconnect voltage
LVD	low-voltage disconnect
LVDH	low-voltage disconnect hysteresis
MA	methylammonium
MBE	molecular beam epitaxy
MEG	multiple exciton generation
MG-Si	metallurgical-grade silicon
MOCVD	metal-organic chemical vapour deposition
MPP	maximum power point

MPPT	maximum power point tracking (tracker)
nc-Si:H	hydrogenated nanocrystalline silicon
NOCT	nominal operating cell temperature
P&O	perturb and observe
PBT	payback time
PECVD	plasma-enhanced chemical vapour deposition
PERC	passivated emitter and rear cell
PERL	passivated emitter, rear locally diffused
PID	potential-induced degradation
PSG	phosphosilicate glass
PSH	peak sun hours
PV	photovoltaic
PV/T	photovoltaic-thermal (hybrid collector)
PWM	pulse width modulation
SHS	solar home system
SOC	state of charge
SPV	solar photovoltaic
SPWM	sinusoidal pulse width modulation
SQ	Shockley-Queisser
SREDA	Sustainable and Renewable Energy Development Authority (Bangladesh)
SRH	Shockley-Read-Hall (trap-assisted recombination)
STC	standard test conditions (1000 W/m^2, AM1.5, 25 °C)
TCO	transparent conducting oxide
TCS	trichlorosilane ($HSiCl_3$)
THD	total harmonic distortion
TOPCon	tunnel oxide passivated contact
VA	volt-ampere
VB	valence band
VFD	variable frequency drive

VR	voltage regulation (set point)
VRH	voltage regulation hysteresis
Wp	watt-peak (power at STC)

Built from the scanned ESE 3207 papers (2019, 2020, 2021, 2022, 2023). Question text is cleaned up from the scans.